

Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

Amaze Yourself!

Read Electronics For You

You'll realise how much you didn't know, and didn't know that you didn't know!

To Subscribe:
<http://subscribe.efy.in>

Scan This Code



```

rfid_rakeshjaincode
//RAKESH JAIN PROGRAM FOR RFID BASED AUTOMATIC RESTRICTED QUANTITY WATER/FUEL DISPENSER
int count = 0;
char input[12];
int ulnpin1 = 13;
int ulnpin2 = 12;
void setup() {
  Serial.begin(9600);
  pinMode(ulnpin1 , OUTPUT);
  pinMode(ulnpin2 , OUTPUT);
}
void loop() {
  if(Serial.available())
  {
    count = 0;
    while(Serial.available() && count < 12)
    {
      input[count] = Serial.read();
      count++;
      delay(5);
    }
    if(input[0] == 'S' && input[1] == 'A' && input[2] == '0' && input[3] == '0' &&
      input[4] == '1' && input[5] == 'D' && input[6] == 'B' && input[7] == 'C' &&
      input[8] == 'S' && input[9] == 'D' && input[10] == 'A' && input[11] == '6' )
    {
      digitalWrite(ulnpin1 , HIGH);
      digitalWrite(ulnpin2 , HIGH);
      delay(16000);
      digitalWrite(ulnpin1 , LOW);
      digitalWrite(ulnpin2 , LOW);
    }
  }
}

```

Fig. 4: Snippet of the source code

Software

First, obtain the 12-digit unique identification number of the RFID tag on the serial monitor by uploading the source code (see Fig. 2) to Arduino program. Before uploading, connect it to the EM-18 RFID module (MOD2), as shown in Fig. 3.

Put the RFID tag in front of RFID reader module and see its 12-digit decoded value. The decoded values for the author's RFID tags are shown in Table 2.

Next, prepare the code that checks the RFID card number and gives the signal to relay to dispense the liquid according to that RFID card. Here you need to change and set the card ID number according to your RFID card ID number. Fig. 4 shows snippet of the source code.

Circuit and testing

Connect all the components according to circuit diagram shown

TABLE 2 DECODED VALUES FOR RFID TAGS		
RFID Tag	Image	Decoded value from serial monitor (12-digit)
0001948765 029,48221		5A001DBC5DA6
0013946179 212,52547		5900D4CD4303
0013942651 212,49019		5900D4BF7B49
0001956746 029,56202		5A001DDB8A16
0001949492 029,48948		5A001DBF34CC

in Fig. 5.

Whenever an RFID tag is put in front of the RFID reader, both the